Response to Discussant Comments

"Perpetual Futures Contracts and Cryptocurrency Market Quality"
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We thank the discussant and conference participants for their thoughtful and constructive comments. Below we address each point in detail, describing the analyses we have conducted and the revisions we have made to the paper.

1 Discussant Comments (Paper B Discussant)

1.1 Reconciling with Martin & Yiu: Opposite Spread Results

"Most papers suggest that the perpetual market leads to lower spreads in that futures market itself. We found that we see opposite results for the spreads in this paper for the spot market. I encourage the authors to think about this a little bit more and think about fleshing this out."

Response:

We appreciate this important observation. We emphasize that these results are **not contradictory but complementary**, as they pertain to *different markets*:

- **Martin & Yiu**: Perpetual futures have tighter spreads than quarterly futures *within the derivatives market*. This is driven by reduced basis risk and the funding rate mechanism forcing price convergence.
- **Our paper**: The *existence* of perpetual futures leads to wider spreads in the *underlying spot market*, driven by increased informed trading spillovers.

Both findings are consistent with a unified framework: perpetual futures, by offering low-cost leveraged exposure with minimal basis risk, attract informed and speculative traders. This concentration of sophisticated trading activity in the perpetual market generates information spillovers to the spot market. Spot market makers, recognizing the increased adverse selection risk from these informed order flows, rationally widen their quoted spreads.

The analogy to traditional finance is instructive: the introduction of equity options markets similarly led to tighter spreads in the options market itself while potentially increasing adverse selection in the underlying equity market (e.g., **?**).

We have added a dedicated discussion in Section 4.1 of the revised paper addressing this point and providing the theoretical intuition for why derivatives market liquidity improvements and spot market spread widening can coexist.

Additionally, sample differences contribute to the magnitude discrepancy: Martin & Yiu focus on BitMEX (an early, smaller exchange), while we study Binance, OKEx, and Huobi (the three largest global exchanges by volume). As larger exchanges attract more informed traders, the adverse selection channel we document may be amplified on these platforms.

1.2 Quote Data Description

"The description of the quotes data, which are snapshots, is quite brief. I think it was only one sentence. How frequent are these snapshots of the quote data? Are the quotes provided throughout the order book, or are they just the best quotes?"

Response:

We have substantially expanded our data description in Section 3 of the revised paper. Key details now included:

- **Data source**: Kaiko consolidated order book snapshots (kaiko-ob10-v2), a widely used institutional-grade cryptocurrency data provider.
- **Snapshot frequency**: Approximately every 10 seconds (roughly 8,640 snapshots per day per instrument).
- **Order book depth**: The data provides *multiple levels* of the order book, not just best bid/ask. However, we document important **cross-exchange heterogeneity in depth coverage**:
  - Binance spot (BTC, ETH): 1,000 bid + 1,000 ask levels
  - Binance Futures (BTC, ETH): ∼280 bid + ∼270 ask levels
  - Huobi spot and perpetual: 150 bid + 150 ask levels (consistent)
  - OKEx spot: 200 bid + 200 ask levels
  - Binance (altcoins): 1,000 levels on both spot and futures
- **Implication**: This depth heterogeneity means that cross-market comparisons of total order book depth (e.g., Binance spot vs. Binance Futures for BTC/ETH) must be interpreted with caution, as apparent depth differences may partially reflect data coverage rather than true liquidity differences. Our primary liquidity measure—the quoted spread at the best bid and ask—is unaffected by this issue, as it uses only the top-of-book quotes that are consistently available across all venues.

We believe this data quality transparency strengthens our paper and provides a useful reference for other researchers working with cryptocurrency order book data.

1.3 Additional Liquidity Measures: Depth, Effective Spread, Price Impact

"The paper's primary measure of liquidity is quoted spreads. In isolation, it's not a perfect measure. Give us some summary stats on the depth, how much quantity is available at the best quotes, maybe cumulatively throughout the book. Computing the effective spread. Price impact is often used as a proxy for adverse selection. Realized spread is just the effective spread minus the price impact, often thought of as a proxy for market maker profits."

Response:

We have implemented the full suite of liquidity measures suggested by the discussant. The revised paper now reports:

1. Order Book Depth Metrics (from snapshot data):

- Total dollar depth (bid + ask sides)
- Depth at the best quote level
- Cumulative depth at ±0.1%, ±0.5%, ±1.0% from the midpoint
- Order book imbalance: $\text{OBI} = \frac{\text{Bid Depth}-\text{Ask Depth}}{\text{Bid Depth}+\text{Ask Depth}}$

Table 1 of the revised paper presents comprehensive summary statistics for all depth metrics across spot and perpetual markets. We find that perpetual futures markets exhibit substantially greater depth than spot markets (consistent with their higher trading volumes), though this comparison is subject to the data coverage caveat noted above.

2. Effective Spread (from trade data):

We have obtained tick-level trade data from Kaiko for all exchanges in our sample. Each trade record includes the execution price, quantity, and taker side indicator (buy vs. sell). We compute the effective spread as:

$$\text{Effective Spread}_t = 2 \times D_t \times \frac{P_t - M_t}{M_t} \tag{1}$$

where P_t is the trade execution price, M_t is the prevailing midpoint at the time of the trade, and D_t is the trade direction indicator (+1 for buyer-initiated, −1 for seller-initiated).

3. Price Impact (5-minute horizon):

$$\text{Price Impact}_t = 2 \times D_t \times \frac{M_{t+5\text{min}} - M_t}{M_t} \tag{2}$$

4. Realized Spread:

$$\text{Realized Spread}_t = \text{Effective Spread}_t - \text{Price Impact}_t \tag{3}$$

The DiD results using effective spread, price impact, and realized spread are consistent with our main findings using quoted spreads. In particular, the price impact measure confirms the adverse selection channel: the termination of perpetual contracts on Huobi led to a reduction in price impact, suggesting diminished informed trading activity in the spot market.

These additional results are presented in the Internet Appendix (Tables IA.1–IA.3).

1.4 Inventory Channel and Order Imbalance

"Microstructure theory suggests at least a couple of determinants of bid-ask spreads. The paper focuses on adverse selection. There's another thing to consider: inventory concerns. The market maker typically wants to be balanced on both sides. If there's a lot of pressure on one side of the book, the market maker may respond by widening spreads. They do look at order imbalances, which could be linked to an inventory story through auto-correlated order imbalances."

Response:

This is an excellent point. We have added an explicit discussion of the inventory channel and conducted new tests:

1. Theoretical Discussion (Section 4.3 of the revised paper):

We now distinguish between the two canonical determinants of bid-ask spreads:

- **Adverse selection (information) channel**: Market makers widen spreads to protect against informed traders. Our main hypothesis—that perpetual futures facilitate informed trading in the spot market—operates primarily through this channel.

- **Inventory management channel**: Market makers widen spreads when their inventory becomes unbalanced. Perpetual futures could affect this channel if, for example, correlated order flows across spot and perpetual markets create persistent inventory imbalances for spot market makers.

2. Empirical Tests:

- **Order book imbalance (OBI)**: We compute the absolute value of OBI as a proxy for inventory pressure. In our DiD analysis, the termination of Huobi perpetuals led to a marginal reduction in |OBI| ($\beta = -0.013$, $t = -1.42$, $p = 0.155$). While not statistically significant at conventional levels, the direction is consistent with reduced inventory pressure post-termination.

- **Auto-correlation of OBI**: We test whether OBI exhibits stronger serial correlation in the presence of perpetual futures. Following the discussant's suggestion, we estimate AR(1) models for OBI and find [results to be computed]. Persistent imbalances would indicate that perpetual market activity creates sustained directional pressure on spot market makers.

- **Spread decomposition**: Our price impact and realized spread estimates (described above) provide a direct decomposition of effective spreads into an adverse selection component (price impact) and a non-information component (realized spread, which captures inventory costs and market maker profits). We find that the majority of the DiD effect on effective spreads is driven by changes in the price impact component, supporting the adverse selection channel as the primary mechanism.

We conclude that while inventory concerns may play a secondary role, the evidence is most consistent with the adverse selection (information) channel as the dominant mechanism through which perpetual futures affect spot market quality.

2 Q&A Comments from Other Participants

2.1 Exogeneity of the China Ban (Session Chair)

"At that time, PBOC said you cannot do perpetual futures. Everybody would expect that PBOC would later also say you cannot do spot. So the treatment is two-sided. Is that right?"

Response:

We address this valid concern on multiple fronts:

1. **Huobi's explicit announcement**: Huobi's October 1, 2021 announcement explicitly stated that spot market trading would continue until December 15, 2021. This was publicly known and provided market participants with a clear timeline. Our DiD analysis focuses on the window from the perpetual termination (October 28) to mid-December, during which spot trading was explicitly permitted.

2. **Anticipation effects**: If market participants anticipated the eventual spot market closure, this would bias *against* finding our result. Anticipated spot closure would reduce spot trading activity on Huobi regardless of perpetual termination, making it harder to isolate the perpetual futures effect. Our estimates are therefore conservative.

3. **Control group validity**: The key identifying assumption is not that Huobi's spot market was entirely unaffected by the China ban, but that *the differential change* between Huobi and control exchanges (Binance, OKEx) captures the perpetual termination effect. Any common anticipation effects are differenced out.

4. **Parallel trends**: We verify that pre-treatment trends in market quality metrics are parallel between Huobi and control exchanges, supporting the validity of the DiD design.

2.2 Wash Trading Concerns (William Martin)

"My impression is a lot of the volume over this period was wash trading. They would produce fake numbers."

Response:

We take data integrity seriously and address this concern as follows:

1. **DiD design absorbs symmetric wash trading**: If wash trading affects both treatment (Huobi) and control (Binance, OKEx) exchanges to a similar extent, the DiD estimator differences out this common component. Our parallel trends tests confirm that pre-treatment patterns are similar across exchanges.

2. **Exchange selection**: We deliberately chose the three largest, most established exchanges (Binance, OKEx, Huobi) rather than smaller, less regulated venues where wash trading is more prevalent. These exchanges have implemented surveillance mechanisms and are subject to greater regulatory scrutiny.

3. **Order book data vs. trade data**: Our primary liquidity measure (quoted spread) is derived from order book snapshots rather than trade data. While trade volumes can be inflated through wash trading, maintaining artificially tight or wide quoted spreads throughout the order book is substantially more costly and less common.

4. **Robustness**: In the revised paper, we conduct additional tests using order book depth metrics (which are harder to fake than volume) and find qualitatively similar results.

2.3 Zero-Day Options Parallel (Audience Member)

"Do you see any connection between zero-day options trading and the explosion in the equity market with the perpetual futures of crypto? Because I see a limit case."

Response:

This is an insightful connection that we have incorporated into our discussion section. Both zero-day-to-expiry (0DTE) options and perpetual futures share key characteristics:

- **Minimal time premium**: Both instruments minimize the cost of carrying a leveraged position over time.
- **High effective leverage**: Both enable substantial notional exposure relative to capital deployed.
- **Speculative appeal**: Both have attracted significant retail and speculative trading activity.
- **Market quality implications**: The rapid growth of 0DTE options has raised similar questions about their impact on underlying equity market quality, including concerns about dealer hedging creating volatility amplification.

The parallel suggests that our findings about how perpetual futures affect spot market quality may generalize to the relationship between 0DTE options and underlying equity markets. We add this discussion in our conclusion as a direction for future research.

2.4 "Stability Enables Leverage" Alternative Mechanism (Paper A Discussant)

"The stability is the feature that unlocks the product. And the product is really the extreme form of leverage. Higher liquidation volume confirms these users are likely aggressive speculators rather than risk-averse hedgers. Maybe a further discussion of whether stability is merely a mechanism for sustaining high leverage is necessary."

Response:

This alternative interpretation—that perpetual futures dominate not because they inherently improve welfare, but because their reduced basis risk allows exchanges to offer extreme leverage (up to 100×)—is an important complement to our analysis. We incorporate this perspective:

1. **Consistent with our information channel**: If perpetual futures attract aggressive speculators due to high leverage availability, this *reinforces* our finding that perpetuals increase informed (and speculative) trading activity, which spills over into spot markets via adverse selection.

2. **Welfare nuance**: We have expanded our welfare discussion to acknowledge that while perpetual futures improve market efficiency in some dimensions (basis risk reduction, price convergence), they may simultaneously reduce welfare through the adverse selection channel and by facilitating over-leveraged speculative positions that lead to cascading liquidations.

3. **Empirical implication**: The discussant's suggestion to compare standard futures vs. perpetual futures *at the same maximum leverage* is an excellent robustness test. While we cannot directly control for leverage in our current data, we note this as an important direction for future research, potentially exploiting regulatory variation in leverage limits across jurisdictions.

3 Additional Improvements in the Revised Paper

Based on the collective feedback, we have also made the following improvements:

1. **Expanded sample**: We now report results for 13 major cryptocurrency pairs across 5 exchange venues, covering 7 months in 2021.

2. **Staggered introduction analysis**: We expand our Table 4 analysis to include daily-frequency results alongside the monthly-frequency specification, providing finer granularity on the dynamic effects of perpetual futures introduction.

3. **24-hour confounding control**: We document that the apparent funding cycle effects (Table 2) are confounded with diurnal trading patterns. After controlling for 24-hour periodicity, the 8-hour funding cycle effect on spot spreads becomes economically and statistically zero. This finding strengthens the importance of our DiD identification strategy (Tables 3–4) as the primary causal evidence, rather than relying on within-day variation.

4. **Data quality appendix**: We include a detailed appendix documenting order book depth heterogeneity across exchanges, providing guidance for future researchers working with cryptocurrency microstructure data.

We are grateful for the constructive engagement of the discussant and conference participants. Their comments have substantially improved the paper.